

Homework 2 : Orthanc

Total points: 130

Learning Objectives:

In this assignment, you will work hands-on with DICOM data using an open-source DICOM server. You will learn how to use some common Python libraries for medical imaging and DICOM data and how to modify the data on the server. You are expected to know the basics of Python as a prerequisite to complete this assignment.

Instructions:

This assignment consists of four parts and has **two** submission placeholders on Canvas:

1. Install a local instance of Orthanc and upload DICOM data to it.
2. Investigate the DICOM data.
3. Modify the DICOM data.
4. Explain your approach and submit the report **via Homework 2-Report** assignment.

Part 1: Installing Orthanc and Uploading Data

1. Visit <https://www.orthanc-server.com/download.php> to download the latest version of Orthanc for your OS (Windows, Mac, or Linux), and then install it on your local machine.
2. Visit The Cancer Imaging Archive (TCIA) at <https://www.cancerimagingarchive.net/> and download the chest CT examination for patient "A034518". In TCIA, it's the study called CT ABD PELVIS(WITH CHEST IMAGES) W IV CON. The full study is a little under 700MB.
 - a. You will need the NBIA Data Retriever to download the study. Instructions can be found [at this link](#):
3. Run Orthanc and upload the dataset you downloaded from TCIA.
4. Upload the '*sample-cxr.dcm*' de-identified chest radiograph provided as part of this assignment.

Before you start writing code, review the DICOM headers and familiarize yourself with the fields and the data they contain.

Part 2: Investigate the DICOM Data

You will be expected to write a Python script to query for and retrieve the CT examination and analyze the DICOM header to answer a set of questions. You must use the **pyorthanc** library, otherwise the autograder might not be able to run your code.

Use DICOM transactions to perform the following steps for the CT examination:

1. Use the following parameters to connect to the Orthanc server:
 - IP address: localhost
 - AE title: ORTHANC
 - Port number: 4242
2. Query for the patient with ID "A034518" and retrieve their study.
3. Write code to answer the following questions.
 - a. What is the patient's age? (hint: this is found as a string)
 - b. What is the patient's sex?
 - c. What kind of study did the patient have?
 - d. What modality (i.e., type of imaging such as XR, CT, MRI) is this study?
 - e. Who is the manufacturer of the scanner the study was performed on?
 - f. What is the patient ID?
 - g. How many series are there in the study?
 - h. What's the study instance UID?
 - i. For image 130 of series 4,
 - i. How many rows does it have?
 - ii. How many columns does it have?
 - iii. What are the minimum, maximum, and mean pixel values of this image?
4. Your code should save the above data in a json file with the file name **"study_info.json"**. The json format must follow the format below. Please note that **"xxx"** are string values and **###** are numeric.

```
{
  "Age": "xxx",
  "Sex": "xxx",
  "StudyDescription": "xxx"
  "Modality": "xxx",
  "Manufacturer": "xxx",
  "PatientID": "xxx",
  "NumSeries": ###,
  "StudyInstanceUID": "xxx",
  "NumRows": ###,
  "NumCols": ###,
```

```
"MinPixelVal": ###,  
"MaxPixelVal": ###,  
"MeanPixelVal": ###  
}
```

Part 3: Modify the DICOM Data

You will be expected to enhance your Python script to do the following:

1. Queries for and retrieves the CXR examination, modifies the DICOM header, and sends it back to Orthanc.

Use DICOM transactions to perform the following steps for the chest radiograph:

1. Remain connected to the Orthanc server.
2. Query for the study with patient ID 3142537564 and retrieve their study.
3. Write code to modify the following fields:
 - a. Change the study instance UID to [1.2.840.20252025](#)
 - b. Change the study accession number to [EAS5850-12345678](#)
 - c. Change the date of birth to [January 1, 1978](#)
 - d. Change the patient sex to [O](#) (the letter, not zero)
 - e. Change the date on which the exam was performed to [December 31, 2022](#)
 - f. Change the patient's ID to [8675309](#)
 - g. Change the name of the referring physician who ordered the study to [Thomas Spock](#)
4. Update the chest radiograph on the Orthanc server with the previous values.

Make sure you format any fields you modify to match the formatting in the original DICOM files. Do not change any other values in the DICOM header or the pixel data other than the ones we ask you to change.

Part 4: Explain Your Approach

Write up your approach to this assignment using a maximum of one page, single-spaced.

1. Summarize your approach to the assignment.
2. How is working with DICOM images different from working with other types of images?
3. What accommodations, if any, did you have to make for this in your code?
4. Which part of the assignment created the highest number of errors for you and how did you handle them? Give at least one concrete example with code snippets.

Save your report file with the name `pennkey_hw2.pdf`. Your report must be a PDF. It will have a separate Canvas / Gradescope submission from your code.

Submission:

Reminder: There are two required submissions for Homework 2 on Canvas / Gradescope.

1. Submission Part 1 (Homework 2 on Gradescope) - Python code (autograded)
 - o Submit your code as a Python (`.py`) file. Notebook (`.ipynb`) files will not work with the autograder. If you plan to use a notebook on a platform like Google Colab to do the assignment, convert it to a Python file before uploading.
 - o Your Python file name must be of the format `pennkey_hw2.py`, otherwise, it will not be run by the autograder. Replace `pennkey` in the file name with your pennkey (username)
 - o You do not need to import any packages other than `pyorthanc` and `json`. You may use `numpy` if needed. If you import packages that are not included in the grading environment, your program will fail.
 - o You may submit to the autograder as many times as you want before the assignment deadline. There are no hidden or additional post-submission tests. Any late submission may result in point deductions.
2. Submission Part 2 (Homework 2 - Report on Gradescope) - Report (manually graded)
 - o Submit your report as a PDF (`.pdf`) file.
 - o Your report file name must be of the format `pennkey_hw2.pdf`.
 - o You may submit your report multiple times, but only the latest version will be manually graded after the submission deadline.